

Every face of $\Sigma(P, Q) \subseteq \mathbb{R}^p$ is defined by some linear function on $\mathbb{R}^p$. Thus it defines a map $\pi^{\mathbf{c}} : P \rightarrow Q^{\mathbf{c}}$, and thus a π -coherent subdivision $\mathcal{F}^{\mathbf{c}}$ of Q . Again a simple computation shows that for a continuous section $\gamma : Q \rightarrow P$, the point $\frac{1}{\text{vol}(Q)} \int \gamma d\mathbf{x} \in \Sigma(P, Q)$ lies in the face defined by $\mathbf{c}$ if and only if the image of the section is entirely contained in the collection of faces $\mathcal{F}^{\mathbf{c}} \subseteq L(P)$.

This yields a bijection between the faces of $\Sigma(P, Q)$ and the coherent subdivisions of Q , and thus in particular, between the *coarsest* such subdivisions and the *facets* of $\Sigma(P, Q)$. $\square$

The correspondence between faces of $\Sigma(P, Q)$ and π -coherent subdivisions of Q also yields an explicit method to construct the vertices and the facets of $\Sigma(P, Q)$, which we will use heavily.

Namely, for every vertex there is a unique section, which we only have to integrate. For the facets we always proceed as follows. For every coarsest subdivision of $Q \subseteq \mathbb{R}^q$ that has a chance to be π -coherent, we construct a “lifting” $\widehat{Q} \subseteq \mathbb{R}^{q+1}$ such that the subdivision comes from the lower faces of $\widehat{Q}$. If we can find the affine map $\pi^{\mathbf{c}} : P \rightarrow \widehat{Q}$, then the function $\mathbf{c}$ that defines the facet can be reconstructed via $\mathbf{c}\mathbf{x} = (\pi^{\mathbf{c}}(\mathbf{x}))_{q+1}$.

9.2 Some Examples

Some special cases of the fiber polytope construction are easily analyzed.

If $Q = \{\mathbf{q}\}$ is a point ($q = 0$), then $\Sigma(P, Q) = \Sigma(P, \{\mathbf{q}\}) = P$.

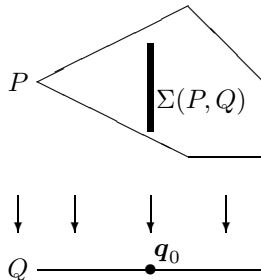
More generally, if $P = R \times Q$ is a product, and $\pi : R \times Q \rightarrow Q$ is the canonical projection, then it is easy to see that $\Sigma(P, Q) = \Sigma(R \times Q, Q) \cong R$: the fiber polytope is a translate of R . In fact, we get

$$\Sigma(R \times Q, Q) = R \times \{\mathbf{q}_0\},$$

where $\mathbf{q}_0$ denotes the barycenter of Q .

If $p = q$, then we have $P = Q$ and $\Sigma(Q, Q) = \{\mathbf{q}_0\}$.

If $p = q + 1$, then $\Sigma(P, Q)$ is an edge $[\mathbf{s}^\downarrow, \mathbf{s}^\uparrow]$, which π maps to the barycenter of Q .



The lower end $\mathbf{s}^\downarrow$ of the interval arises as the integral of the collection $\mathcal{L}^\downarrow$ of “lower faces” of P , while $\mathbf{s}^\uparrow$ is the integral of the collection of “upper faces,” divided by $\text{vol}(Q)$ in both cases.

These were trivial cases — we will now do two more interesting ones. We start with the permutahedron, and the (more general) “monotone path polytopes.” Then we will construct the associahedron, as the “secondary polytope” of an n -gon. So both examples are special cases of important constructions.

Definition 9.7. Let $P \subseteq \mathbb{R}^p$ be a p -dimensional polytope, and consider a nonzero linear function $\mathbf{a} \in (\mathbb{R}^p)^*$ on P . This defines a projection

$$P \longrightarrow Q := \{\mathbf{a}\mathbf{x} : \mathbf{x} \in P\} \subseteq \mathbb{R}^1$$

to the 1-dimensional polytope $Q = [a_{\min}, a_{\max}]$, where

$$a_{\min} = \min_{\mathbf{x} \in P} \mathbf{a}\mathbf{x}, \quad a_{\max} = \max_{\mathbf{x} \in P} \mathbf{a}\mathbf{x}.$$

The fiber polytope of this projection,

$$\Pi(P, \mathbf{a}) := \Sigma(P, \{\mathbf{a}\mathbf{x} : \mathbf{x} \in P\})$$

is the *monotone path polytope* of P and $\mathbf{a}$.

By Theorem 9.6, the vertices of $\Pi(P, \mathbf{a})$ are in bijection to certain paths on the boundary of P that are monotone (strictly increasing) with respect to the function $\mathbf{a}\mathbf{x}$. In fact, every path

$$\phi: \quad \mathbf{v}_0 \longrightarrow \mathbf{v}_1 \longrightarrow \dots \longrightarrow \mathbf{v}_{n-1} \longrightarrow \mathbf{v}_n,$$

for vertices $\mathbf{v}_i \in P$ with

$$a_{\min} = \mathbf{a}\mathbf{v}_0 < \mathbf{a}\mathbf{v}_1 < \dots < \mathbf{a}\mathbf{v}_{n-1} < \mathbf{a}\mathbf{v}_n = a_{\max},$$

defines a section

$$\begin{aligned} \gamma^\phi: Q &\longrightarrow P, \\ x &\longmapsto t \cdot \mathbf{v}_{i-1} + (1-t) \cdot \mathbf{v}_i \\ &\quad \text{for } x = t \cdot \mathbf{a}\mathbf{v}_{i-1} + (1-t) \cdot \mathbf{a}\mathbf{v}_i \quad (0 \leq t \leq 1). \end{aligned}$$

Every such section γ^ϕ defines a point in the monotone path polytope $\Pi(P, \mathbf{a})$, namely the integral

$$\begin{aligned} \mathbf{v}^\phi &= \frac{1}{\text{vol}(Q)} \int_Q \gamma^\phi(x) dx = \\ &= \frac{1}{a_{\max} - a_{\min}} \left((\mathbf{a}\mathbf{v}_1 - \mathbf{a}\mathbf{v}_0) \frac{\mathbf{v}_0 + \mathbf{v}_1}{2} + \dots \right. \\ &\quad \left. \dots + (\mathbf{a}\mathbf{v}_n - \mathbf{a}\mathbf{v}_{n-1}) \frac{\mathbf{v}_{n-1} + \mathbf{v}_n}{2} \right). \end{aligned}$$

Not every such point $\mathbf{v}^\phi$ is a vertex of the monotone path polytope. For this it is necessary that all the segments $[\mathbf{v}_{i-1}, \mathbf{v}_i]$ are edges of P , and

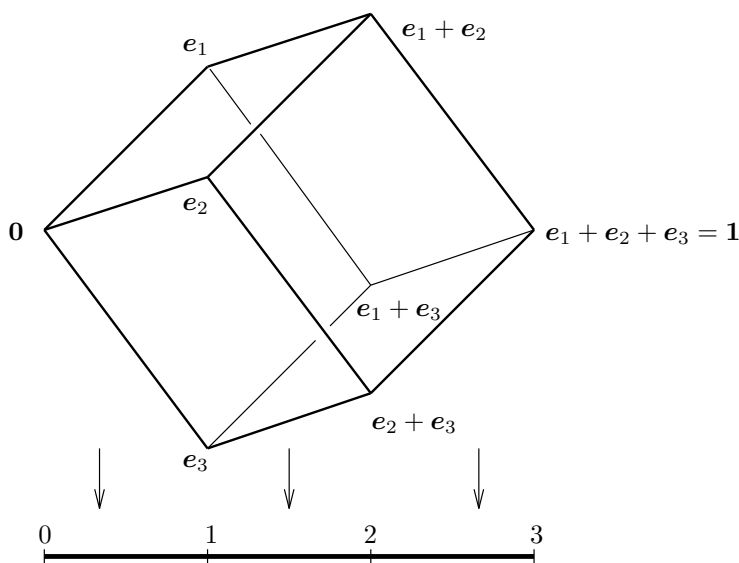
ϕ has to be a monotone edge path on P which is “selected” by a secondary objective function $\mathbf{c}$. Now if you disentangle definitions, this means that the path defines a coherent section (that is, $\mathbf{v}^\phi$ is a vertex of $\Sigma(P, \mathbf{a})$) if and only if ϕ is a path that could occur under Borgwardt’s [125] *shadow-vertex algorithm*. This is a very natural pivot rule for linear programming that is also used under the name “Gass-Saaty rule” in parametric optimization; see Klee & Kleinschmidt [328].

The following example describes a very special case, in which we will again meet our friend from Lecture 0 (Example 0.10): the permutahedron.

Example 9.8 (Permutahedron). [78, Ex. 5.4]

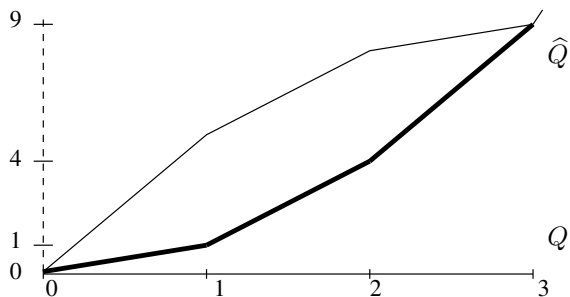
Let $P = [0, 1]^n \subseteq \mathbb{R}^n$ be the unit cube in $\mathbb{R}^n$, and let $Q = [0, n]$ be a segment in $\mathbb{R}^1$. Then we get a projection

$$\pi : [0, 1]^n \longrightarrow [0, n], \quad \mathbf{x} \longmapsto \mathbb{1} \mathbf{x} = \sum_{i=1}^n x_i.$$



The vertices of $\Pi([0, 1]^n, \mathbb{1}) = \Sigma([0, 1]^n, [0, n])$ correspond to increasing edge paths. Here we have the very special situation that all those paths have their vertices at the same values of the linear function, $\mathbb{1} \mathbf{v}_i = i$, and thus they all induce the same subdivision of Q , which breaks $[0, n]$ into the segments $[i-1, i]$. These finest subdivisions arise from maps $[0, 1]^n \longrightarrow \widehat{Q}$, $\mathbf{x} \longmapsto \begin{pmatrix} \pi(\mathbf{x}) \\ \mathbb{1} \mathbf{x} \end{pmatrix}$ such that $\widehat{Q}$ is a convex $2n$ -gon. We may choose the lower vertices of $\widehat{Q}$ to lie on the curve $f(k) = k^2$. Now $\widehat{Q}$ is a projection of the cube $[0, 1]^n$, so we get $\widehat{Q}$ to be the centrally symmetric, convex $2n$ -gon

$$\text{conv} \left(\left\{ \begin{pmatrix} i \\ i^2 \end{pmatrix} : i = 0, 1, 2, \dots, n \right\} \cup \left\{ \begin{pmatrix} i \\ n^2 - (i-n)^2 \end{pmatrix} : i = 0, 1, 2, \dots, n \right\} \right).$$



The possible projection maps $\hat{\pi} : P \longrightarrow \hat{Q}$ correspond to permutations: the permutation $\sigma = \sigma(1)\sigma(2)\dots\sigma(n)$ corresponds to the map

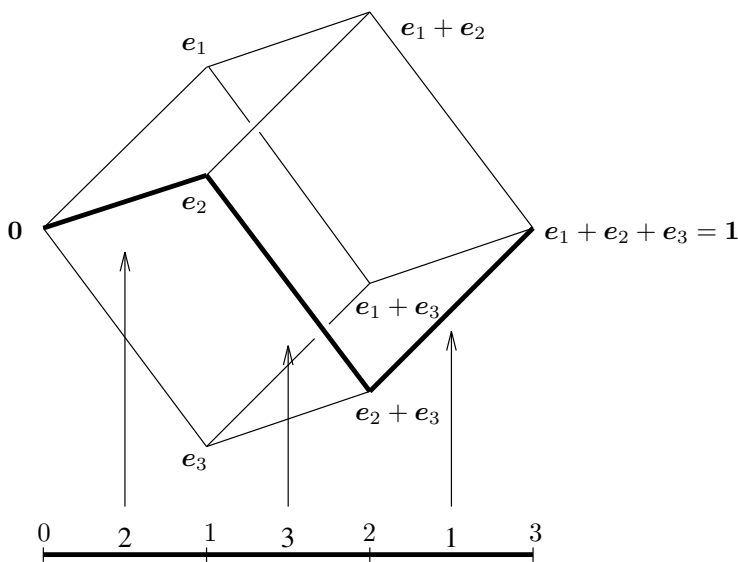
$$\hat{\pi}^\sigma = \left(\begin{smallmatrix} \pi \\ \mathbf{c}^\sigma \end{smallmatrix} \right) : \mathbf{e}_{\sigma(i)} \longmapsto \binom{i}{i^2} - \binom{i-1}{(i-1)^2} = \binom{1}{2i-1},$$

which maps $\mathbf{e}_{\sigma(1)} + \dots + \mathbf{e}_{\sigma(i)} \longmapsto \binom{i}{i^2}$. From this we derive $\mathbf{c}^\sigma \mathbf{e}_{\sigma(i)} = i^2 - (i-1)^2 = 2i-1$, and thus recover the linear function $\mathbf{c}^\sigma \in (\mathbb{R}^n)^*$ as

$$\mathbf{c}^\sigma \mathbf{x} = \sum_{k=1}^n (2k-1)x_{\sigma(k)} = \sum_{k=1}^n (2\sigma^{-1}(k)-1)x_k.$$

The (unique) section $\gamma^\sigma : [0, n] \longrightarrow [0, 1]^n$ that minimizes the integral $\int_0^n \mathbf{c}^\sigma \gamma(x) dx$ describes a path in the 1-skeleton of $[0, 1]^n$:

$$\gamma^\sigma : \mathbf{0} \longrightarrow \mathbf{e}_{\sigma(1)} \longrightarrow \mathbf{e}_{\sigma(1)} + \mathbf{e}_{\sigma(2)} \longrightarrow \dots \longrightarrow \mathbf{e}_{\sigma(1)} + \dots + \mathbf{e}_{\sigma(n)} = \mathbf{1}.$$



For example, our drawing illustrates the path on $P = [0, 1]^3$ corresponding to $\sigma = 231$, which is selected by $\mathbf{c}\mathbf{x} = x_2 + 3x_3 + 5x_1$.

Note that this is a special case of our discussion and computation after Definition 9.7. Here the integral of γ^σ is given by the sum

$$\begin{aligned}
 \int_0^n \gamma^\sigma(\mathbf{x}) d\mathbf{x} &= \frac{1}{2} \left((\gamma^\sigma(0) + \gamma^\sigma(1)) + (\gamma^\sigma(1) + \gamma^\sigma(2)) + \dots \right. \\
 &\quad \left. \dots + (\gamma^\sigma(n-1) + \gamma^\sigma(n)) \right) \\
 &= \frac{1}{2} \left((2n-1)\mathbf{e}_{\sigma(1)} + (2n-3)\mathbf{e}_{\sigma(2)} + \dots + (1)\mathbf{e}_{\sigma(n)} \right) \\
 &= \frac{1}{2} \sum_{i=1}^n (2n+1-2i)\mathbf{e}_{\sigma(i)} \\
 &= \frac{2n+1}{2} \mathbf{1} - \begin{pmatrix} \sigma^{-1}(1) \\ \vdots \\ \sigma^{-1}(n) \end{pmatrix}.
 \end{aligned}$$

We have to divide this by $\text{vol}[0, n] = n$ to get the vertices of the fiber polytope. Thus the fiber polytope of the projection π is an affine image, under

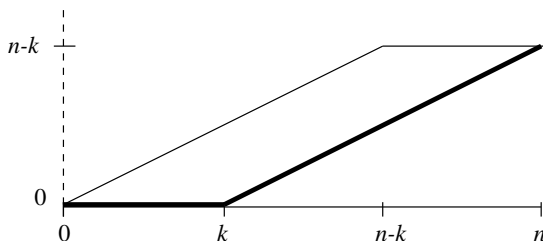
$$\mathbf{x} \mapsto \left(1 + \frac{1}{2n}\right)\mathbf{1} - \frac{1}{n}\mathbf{x},$$

of the “usual” representation of the permutahedron, which represents the permutation σ by the column vector whose entries are given by σ^{-1} :

$$\Pi([0, 1]^n, \mathbf{1}) = \Sigma([0, 1]^n, [0, n]) = \left(1 + \frac{1}{2n}\right)\mathbf{1} - \frac{1}{n}\Pi_{n-1} \cong \Pi_{n-1}.$$

To derive inequalities for the facets of this fiber polytope, we consider the coarsest subdivisions of $[0, n]$, which are generated by functions like $f_k(x) = \max\{0, x - k\}$ for $k = 1, 2, \dots, n-1$. These correspond to the parallelogram

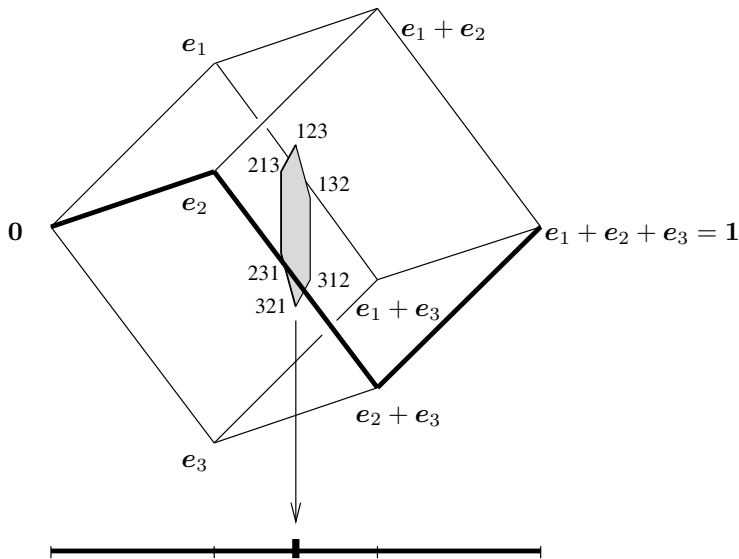
$$\widehat{Q}_k = \text{conv} \left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} k \\ 0 \end{pmatrix}, \begin{pmatrix} n \\ n-k \end{pmatrix}, \begin{pmatrix} n-k \\ n-k \end{pmatrix} \right\}.$$



The corresponding maps $\widehat{\pi} : P \longrightarrow \widehat{Q}_k$ are indexed by the subsets $A \subseteq \{0, \dots, n\}$ with $0 < |A| = n - k < n$, and are given by

$$\widehat{\pi}^A(\mathbf{x}) = \begin{pmatrix} \sum_{i=1}^n x_i \\ \sum_{i \in A} x_i \end{pmatrix} = \begin{pmatrix} \pi(\mathbf{x}) \\ \mathbf{c}^A \mathbf{x} \end{pmatrix}.$$

Our figure illustrates the position of the 2-dimensional permutahedron (a hexagon) as a fiber polytope contained in the 3-cube $[0, 1]^3$. You might notice that the drawing is metrically incorrect — it is just supposed to sketch how vertices of the hexagon and the corresponding monotone paths on the cube are selected by the same linear function on $\mathbb{R}^3$.



Thus we get a complete description of the fiber polytope in terms of equations (because the fiber polytope lies over the barycenter $\frac{n}{2}$ of Q) and inequalities (from coarsest subdivisions) as follows:

$$\begin{aligned} \Sigma([0, 1]^n, [0, n]) &= \{x \in \mathbb{R}^n : \mathbb{1}x = \frac{n}{2}, \\ &\quad \sum_{i \in A} x_i \leq \frac{|A|(2n - |A|)}{2n} \text{ for } \emptyset \subset A \subset [n]\}. \end{aligned}$$

With this we have obtained a complete description of the permutahedron as a fiber polytope. $\square$

We now turn our attention to the secondary polytopes. Our main example here will be Stasheff's associahedron [522], which was first constructed as a polytope by Milnor, Haiman [266] and Lee [355] — see Example 0.10. It turns out that the associahedron can be realized as a fiber polytope $\Sigma(\Delta_n, C_2(n+1))$. For that, we use the existence of canonical maps from an n -simplex to any polytope with $n+1$ vertices.

For the following, we use the special n -simplex

$$\Delta'_n := \text{conv}\{e_i : 0 \leq i \leq n\} \subseteq \mathbb{R}^n,$$

where we set $e_0 := \mathbf{0}$.

Definition 9.9 (Secondary polytopes). [231, 232]

Let $Q \subseteq \mathbb{R}^d$ be a d -polytope with $n + 1$ vertices, $\text{vert}(Q) = \{\mathbf{v}_0, \dots, \mathbf{v}_n\}$. The *secondary polytope* of Q is

$$\Sigma(Q) := (d + 1) \text{vol}(Q) \Sigma(\Delta'_n, Q),$$

where the fiber polytope $\Sigma(\Delta'_n, Q)$ arises from the affine map

$$\pi : \Delta'_n \longrightarrow Q$$

that maps $\mathbb{R}^n \ni \mathbf{e}_i \longmapsto \mathbf{v}_i \in Q$, for $0 \leq i \leq n$.

For every n -simplex $P \subseteq \mathbb{R}^n$ there is a projection map $\pi : P \longrightarrow Q$ that maps the vertices of Δ'_n to the vertices of Q . Furthermore, P and the map π are unique up to affine coordinate changes in $\mathbb{R}^n$. So, for every projection of an n -simplex P to Q , the fiber polytope $\Sigma(P, Q)$ is affinely isomorphic to the secondary polytope $\Sigma(Q)$. (Equivalently, one could also use our “standard” n -simplex $\Delta_n \subseteq \mathbb{R}^{n+1}$.) Thus the secondary polytope is a canonical object associated with any polytope Q . We refer to the papers by Gel’fand, Zelevinsky & Kapranov [232], Billera, Gel’fand & Sturmfels [72], and Billera, Filliman & Sturmfels [71] for extensive discussions. The key observation is that, by Theorem 9.6 together with Exercise 9.4, once the map $\pi : \Delta'_n \longrightarrow Q$ is fixed, every regular subdivision of Q (without new vertices) is π -coherent.

Corollary 9.10. *The vertices of $\Sigma(Q)$ are in bijection with the regular triangulations of Q , via*

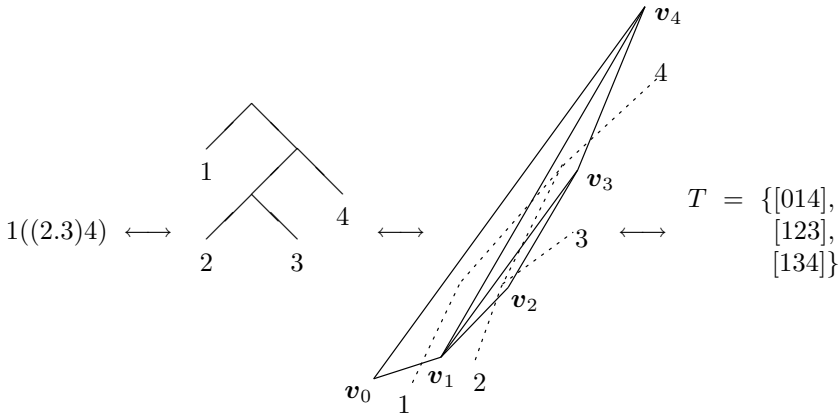
$$T \longleftrightarrow \sum_{[\mathbf{v}_{i_0}, \dots, \mathbf{v}_{i_d}] \in T} \text{vol}[\mathbf{v}_{i_0}, \dots, \mathbf{v}_{i_d}] \cdot (\mathbf{e}_{i_0} + \dots + \mathbf{e}_{i_d}),$$

where every regular triangulation of Q is represented by its collection of d -dimensional simplices.

To get the associahedron from this construction, we use a “well-known” bijective correspondence between the complete bracketings of a string of n letters and the triangulations T without new vertices of the $(n + 1)$ -gon $C_2(n + 1)$. So, to every complete bracketing α of the string $123 \dots n$ of length n we associate the corresponding triangulation of $C_2(n + 1)$, written as a set of triples $T(\alpha) \subseteq \binom{\{0, \dots, n\}}{3}$. We denote by $\mathcal{T}_n$ the set of all these triangulations. For example, we get (using square brackets for the triples)

$$\begin{aligned} \mathcal{T}_3 &= \left\{ \{[013], [123]\}, \{[023], [012]\} \right\}, \\ \mathcal{T}_4 &= \left\{ \{[014], [124], [234]\}, \{[014], [134], [123]\}, \{[024], [012], [234]\}, \right. \\ &\quad \left. \{[034], [013], [123]\}, \{[034], [023], [012]\} \right\}. \end{aligned}$$

Instead of a formal definition, we give a pictorial example that explains the correspondence (in a special case for $n = 4$):



Example 9.11 (Associahedron).

The associahedron (Example 0.10) was first constructed as the secondary polytope of a convex $(n+1)$ -gon by Gel'fand, Zelevinsky & Kapranov [231, Rem. 7c)] [230, Example 7.3.B]. Here we get especially nice coordinates by taking the “cyclic” $(n+1)$ -gon, namely

$$C_2(n+1) = \text{conv}\{\mathbf{v}_i : 0 \leq i \leq n\}, \quad \text{for } \mathbf{v}_i := \begin{pmatrix} i \\ i^2 \end{pmatrix}.$$

In this case the projection map is linear: it maps

$$\mathbf{e}_i \longmapsto \mathbf{v}_i \quad \text{for } 0 \leq i \leq n,$$

and thus in particular $\mathbf{0} = \mathbf{e}_0 \longmapsto \mathbf{v}_0 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$.

The area of a typical triangle spanned by vertices of $C_2(n+1)$ is

$$\text{vol}[\mathbf{v}_i, \mathbf{v}_j, \mathbf{v}_k] = \frac{1}{2}(j-i)(k-i)(k-j)$$

for $i < j < k$, which is an integer. Thus the triangulations T of $C_2(n+1)$ without new vertices are represented by the points

$$\mathbf{v}^T := \sum_{[i,j,k] \in T} \frac{1}{2}(j-i)(k-i)(k-j) \cdot (\mathbf{e}_i + \mathbf{e}_j + \mathbf{e}_k).$$

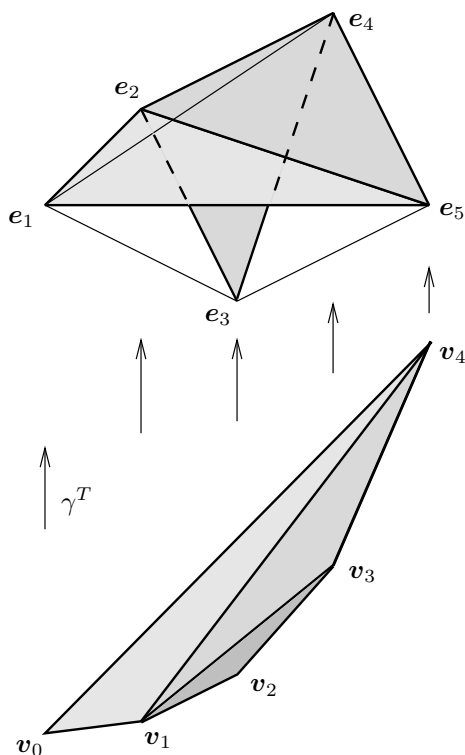
Here the sum is over all triples $i < j < k$ such that $[\pi(\mathbf{e}_i), \pi(\mathbf{e}_j), \pi(\mathbf{e}_k)]$ is a triangle in the triangulation T , of area $\frac{1}{2}(j-i)(k-i)(k-j)$.

Taking this together with the above bijection, this defines a point in $\mathbb{R}^n$ for every complete bracketing of $12 \dots n$. For example, with the bracketing

1((2.3)4) we associate the triangulation $T = \{[014], [123], [134]\}$, compute the volumes $\text{vol}[014] = 6$, $\text{vol}[123] = 1$, $\text{vol}[134] = 3$, and thus we get for this T the point

$$\begin{aligned} \mathbf{v}^T &= 6(\mathbf{e}_0 + \mathbf{e}_1 + \mathbf{e}_4) + 1(\mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3) + 3(\mathbf{e}_1 + \mathbf{e}_3 + \mathbf{e}_4) \\ &= \begin{pmatrix} 6 \\ 0 \\ 0 \\ 6 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 3 \\ 0 \\ 3 \\ 3 \end{pmatrix} = \begin{pmatrix} 10 \\ 1 \\ 4 \\ 9 \end{pmatrix}. \end{aligned}$$

Our figure shows the corresponding section γ^T , whose integral yields $\mathbf{v}^T$.



Let us derive a complete description of the polytope $\Sigma(C_2(n+1))$ in terms of equations and inequalities. For this we first need the volume and the barycenter of $C_2(n+1)$. For this we can use the triangulation $T = \{[012], [023], \dots, [0\ n-1\ n]\}$, and from this compute the volume

$$V_n := \text{vol}(C_2(n+1)) = \sum_{i=2}^n \frac{i(i-1)}{2} = \binom{n+1}{3},$$

and (with an extra computation, see Exercise 9.7) the barycenter

$$\begin{pmatrix} c_n \\ d_n \end{pmatrix} := \mathbf{q}_0(C_2(n+1)) = \begin{pmatrix} \frac{n}{2} \\ \frac{1}{15}(6n^2 + 1) \end{pmatrix} \quad \text{for } n \geq 2.$$

From this we get that $\Sigma(C_2(n+1))$ is contained in the affine subspace of $\mathbb{R}^n$ given by the equations

$$\begin{aligned}\sum_{i=1}^n i x_i &= 3 \operatorname{vol}(C_2(n+1)) c_n = \frac{n^4 - n^2}{4} \\ \sum_{i=1}^n i^2 x_i &= 3 \operatorname{vol}(C_2(n+1)) d_n = \frac{6n^5 - 5n^3 - n}{30}.\end{aligned}$$

How do we get the facet-defining inequalities? We use the method outlined after Theorem 9.6. The facets correspond to the diagonals of $C_2(n+1)$, which we interpret as going from $\mathbf{v}_i$ to $\mathbf{v}_j$, for $0 \leq i < j \leq n$, with $2 \leq j - i < n$. For every such “admissible” pair (i, j) , we construct a regular function

$$f^{ij}\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right) := \max\{0, -y + (i+j)x - ij\}.$$

This formula can be derived from the condition that the linear function $-y + (i+j)x - ij$ vanishes for $\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right) = \left(\begin{smallmatrix} i \\ i^2 \end{smallmatrix}\right)$ and for $\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right) = \left(\begin{smallmatrix} j \\ j^2 \end{smallmatrix}\right)$. The corresponding “lifted polytope” $\widehat{Q}$ is given by

$$\widehat{Q} := \operatorname{conv}\left\{f^{ij}\left(\begin{smallmatrix} k \\ k^2 \end{smallmatrix}\right) : 0 \leq k \leq n\right\}.$$

Since P is a simplex here, we get a canonical map

$$\pi^{\mathbf{c}} : P \longrightarrow \widehat{Q}, \quad \mathbf{e}_k \longmapsto f^{ij}\left(\begin{smallmatrix} k \\ k^2 \end{smallmatrix}\right).$$

Using this, the corresponding facet-defining linear function is $\mathbf{c}^{ij} \in (\mathbb{R}^n)^*$, with

$$\begin{aligned}\mathbf{c}^{ij}\mathbf{x} &= \sum_{k=1}^n f^{ij}\left(\begin{smallmatrix} k \\ k^2 \end{smallmatrix}\right) x_k \\ &= \sum_{k=1}^n \max\{0, -k^2 + (i+j)k - ij\} x_k \\ &= \sum_{k=1}^n \max\{0, (k-i)(j-k)\} x_k \\ &= \sum_{k=i}^j (k-i)(j-k) x_k.\end{aligned}$$

The function $\mathbf{c}^{ij}\mathbf{x}$ will be minimized, by construction, by those vertices $\mathbf{v}^T$ whose triangulation T contains the diagonal (i, j) . One can work out that

the minimum then is

$$\min\{\mathbf{c}^{ij}\mathbf{v}^T : T \in \mathcal{T}_n\} = \binom{j-i+1}{3} \frac{3(j-i)^2 - 2}{10}.$$

Thus we actually get the associahedron,

$$\Sigma(C_2(n+1)) = K_{n-2},$$

and a complete linear description for it, as follows:

$$\begin{aligned} \Sigma(C_2(n+1)) = \left\{ \mathbf{x} \in \mathbb{R}^n : \right. & \sum_{i=1}^n i x_i = \frac{n^4 - n^2}{4} \\ & \sum_{i=1}^n i^2 x_i = \frac{6n^5 - 5n^3 - n}{30}, \\ & \mathbf{c}^{ij}\mathbf{x} \geq \binom{j-i+1}{3} \frac{3(j-i)^2 - 2}{10} \\ & \left. \text{for } 0 \leq i < j \leq n, 1 < j-i < n \right\}. \end{aligned}$$

We can use the PORTA program to check the validity of this description for small n , by inputting the set of vertices, or the inequality system, and checking whether we get a polytope with the correct combinatorics.

Here is an example. For $n = 4$, the linear system above has the form (a file `ass2.ieq` in PORTA input format)

```
DIM = 4

VALID
10 1 4 9

INEQUALITIES_SECTION
1x1 + 2x2 + 3x3 + 4x4 == 60
1x1 + 4x2 + 9x3 +16x4 == 194

      x1                >=  1
      x2                >=  1
      x3                >=  1
2x1+2x2                >= 10
      2x2 +2x3          >= 10

END
```

where the inequalities correspond to the diagonals [02], [13], [24], [03], and [14] (in this order). PORTA requires knowing a valid point for this system, so we give it the point that we had computed before.

Now the PORTA command `traf -v ass2.ieq` produces from this the list of vertices and the incidence matrix for a pentagon, in a file named `ass2.ieq.poi`:

```
DIM = 4

CONV_SECTION
( 1)  1  4  9  6
( 2)  4  1 10  6
( 3)  9  4  1 10
( 4) 10  1  4  9
( 5)  1 10  1  9

END

strong validity table :
\ I      |      |
\ N      |      |
P \ E     |      |
0 \ Q     |  1   |  #
I \ S     |      |
N \       |      |
T \       |      |
S \       |      |
-----
1         | *.*. : 2
2         | *.*. : 2
3         | ..*. : 2
4         | *.*. : 2
5         | *.*. : 2
          | .....
#         | 2222
```

So the polytope is in fact a pentagon, as K_2 should be...

9.3 Constructing the Permuto-Associahedron

We will now describe the construction of the permuto-associahedron of Kapranov [313], as recently achieved in [453] (Example 0.10). There are analogous objects constructed for signed, bracketed permutations in [453], but we do not discuss those here. Also our discussion skips some details: we refer to [453] for the “missing pieces.”

The construction depends on a version of the associahedron in especially nice coordinates. For this, we define

$$\Delta_n^f := \text{conv}\{\mathbf{f}_0, \mathbf{f}_1, \dots, \mathbf{f}_n\}, \qquad \text{for } \mathbf{f}_i := \mathbf{e}_1 + \dots + \mathbf{e}_i, \mathbf{f}_0 = \mathbf{0},$$

as a reference simplex. Again we use the special “cyclic” $(n + 1)$ -gon

$$C_2(n + 1) = \text{conv}\left\{\binom{i}{i^2} : 0 \leq i \leq n\right\}.$$